The Importance of Motivation, Metacognition, and Help Seeking in Web-Based Learning

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Contents

01 Overview

02 Review / Motivation

03 Review / Interest + Metacognition

04 Review / Help Seeking Research

05 Suggestions For Further Research

01

Overview

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前言

This chapter examines whether such **student characteristics as motivation and interest, metacognition**, and **readiness to use help**, such as auxiliary instructional resources, are differentially beneficial for student success in Web-based learning (WBL).

這章節探討**學生的動機**、**後設認知**和**尋求協助**等特徵，對於學習者在網路學習中是否能成功的重要性，並探討這些因素如何影響學習成效。

前言

Duffy and Kirkley (2004a, p. 3) described the rate of WBL development as "**explosive**" echoing the findings of others (NCES, 2003; Shea, Fredericksen, Pickett, & Pelz, 2004). In addition, the government, the military, and agencies in the health and professional sectors, among others, are also increasingly delivering WBL courses. WBL enables students in urban or rural environments who cannot travel to a campus for instruction to receive it. Such convenience is important to caregivers, students, and trainees, as well as to anyone working in unsatisfying positions who needs instruction but is unable to leave home to receive it.

儘管WBL在教育領域的發展迅速，並且在多個行業中已經廣泛應用，可以滿足不同的群體

- 政府、軍方
- 無法面授教學學生

前言

Despite its popularity, relatively little is known empirically about WBL, especially its effectiveness for students with **different characteristics**

在**不同學生特徵**對於網路學習的影響卻缺乏足夠的研究

- 動機、後設認知、尋求協助
- 模糊忍受度
  - 低:傾向立即尋找答案, 當在事件不夠具體或清楚時, 容易產生焦慮感
  - 高:較能接受不立即給予答案, 會更傾向於慢慢探索學習

前言

Phipps and Merisotis (1999) estimated attrition from WBL to be 32%, compared to only 4% in face-to-face settings

WBL的退學率高達32%，而在傳統學習中，退學率僅為4%。

- 對所有學生來說網路學習不一定是理想的學習方式
  - WBL高退學率主要來自於學習過程中的孤立感、自我管理挑戰、缺乏即時支持以及學習效果的不確定性
    - WBL適合自我調制佳的學生
    - WBL面對面交流少、即時反饋較少 —造成→ 學習成效不明顯
    - 傳統課程依賴於同儕互動的學生
    - **場獨立**、場依賴

前言

The attrition and achievement results indicate that WBL is not equally effective for all students and contribute to the widespread agreement among researchers (Bernard et al., 2003; Dillon & Gabbard, 1998; Hartley & Benddixen, 2001; Wisher& Champagne, 2000) that there is an **urgent** need to study the effectiveness of WBL for **different types** of students.

這些退學和成就的數據表明, WBL對所有學生的效果並不一樣。

多位學者認為需要進行**深入研究WBL對不同類型學生的效果**。

Overview

- 討論了動機、後設認知和尋求幫助在網路學習中的重要性，並深入分析了它們如何影響學生的學習表現和成效。

  Sankaran and Bui (2001) found that **motivation** was the **most important** student characteristic related to achievement in WBL courses.

  在WBL當中學生的學習動機，比其他特徵都來的重要

  - 為什麼學習動機是最重要？
    - 學生無法維持持續學習動力，導致進度落後，漸漸退出學習
    - 當遇到挑戰，缺少內在動機，因為這些挑戰所帶來的困難，而選擇放棄
    - 有高動機，才會願意花更多時間去學習，學習效果佳
    - 因WBL缺乏即時回饋，如果有高動機的學習，會自己尋找資源

Overview－Motivation

Motivation, like all affective characteristics, can affect learning only **indirectly** by engaging the **cognitive and metacognitive processes** controlling learning (Tobias, Galvin, & Michna, 2002).

動機**不是直接**決定學習成效的，而是透過動機激發學生的認知（思考過程）和後設認知來間接影響學習成效。

Overview－Motivation

Clarifying the effects of cognitive variables on learning provides an understanding of **how learning occurs**; it is equally important to know why students **choose** and **pursue different learning goals,** that is, to **understand** people's **motivation**, as well as to study the relationships between cognitive and affective variables.

不只是要知道學生怎麼學(他們怎麼學、怎麼記), 了解學生為什麼選擇學習這些知識(學生的動機, 學習目標為何), 這樣的了解可以幫助我們更理解學生學習過程。

Overview－Metacognition

Surprisingly the motivation-cognition or metacognition relationship has received little attention in WBL.

動機對於認知和後設認知學習成效**有間接影響**，但兩者之間關係在WBL的研究當中都沒有被深入探討過

Overview－Metacognition

The scarcity of such research is surprising especially because metacognition was demonstrated (Hill & Hannafin, 1997, p. 38) to be "fundamental to successful Iearning" in environments such as WBL where students work Iargely by themselves, often with requirements that may seem **ambiguous** and are **not** readily clarified, and where learners can easily **lose** their **bearings** (Beasley & Waugh, 1995) and become **disoriented** (Beasley, 1994; Hill, Hannafin, & Land, 1997)by **the flood of information confronting them.**

後設認知對於學習成功是非常重要的，在WBL學習大多獨立學習，學生很容易迷失方向，接收太多大量資訊。

- 為什麼WBL後設認知很重要？

  在學習當中接受太多大量資訊或是任務不明確，導致認知負荷，後設認知非常重要(例如：自我反省、自我調制)來幫助學生更有效處理和調整學習過程。

Overview－Metacognition

Suggestive findings indicate that students with **"undirected" learning strategies**, (Sankaran & Bui, 200l) **learn poorly** from WBL.

當學生在WBL中缺乏明確的學習策略或計劃，他們的學習成效通常會比較差。

Undirected was defined as students having problems processing and coping with large amounts of material and **discriminating** between important and unimportant content. Those results suggest that such students have **ineffective** metacognitive processes such as monitoring of learning, evaluating it, and selecting appropriate learning strategies. Clearly such **metacognitive** processes may be **critical** for success in WBL.

當學生在學習中無法有效分辨，無法準確識別自己學習中的弱點，進而錯過改善學習策略的機會，通常是因為他們缺乏後設認知能力。

Overview－Help Seeking

Students who sought help were **denigrated** as **dependent**, and the act of relying on others was considered the antithesis of what it means to strive for excellence.

學生去尋求幫助，過去很多人會認為這代表學生無法自己解決問題，缺乏自我解決的能力，被視為不夠獨立。

Research has shown that those who seek help are typically more successful in learning than those who do not.

研究顯示，主動尋求幫助的學生通常比那些不尋求幫助的學生學得更成功。

02

Review / Motivation

導讀者：李昀珊

Review / Motivation

It is generally agreed that **motivation initiates, directs,** and **maintains** the **activities controlling learning**.
動機會啟動、引導並維持控制學習的活動。

However, **motivation** must **engage the cognitive and metacognitive processes** controlling learning in order to affect it.
動機必須與認知和後設認知的過程結合，才能對學習產生影響。

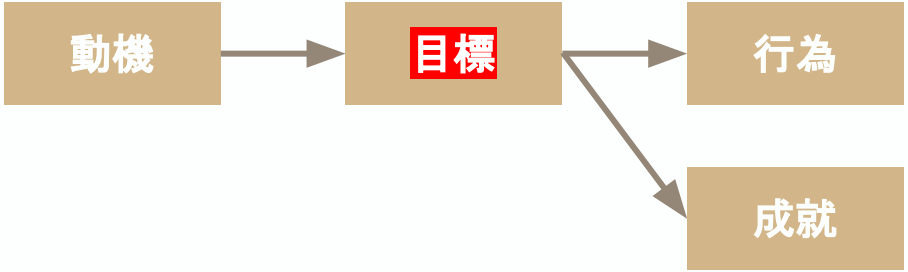
資料來源:參考自《Intrinsic amd Extrinsic Motication》(王淑玲，2024)

Self-regulated learning may be seen as **a combination of metacognitive and motivational processes (Winne & Perry, 2000)**.
自我調制學習 ＝ 後設認知 ＋ 動機過程

Review / Motivational Goal Orientation

Motives are seen as **stable dispositions** to **achieve** or **avoid** specific **outcomes, desires, or wishes** that give rise to the adoption of **various goals** **(Pintrich & Maehr, 2001)**.
動機被視為一種可以達成或避免特定結果、慾望或願望的穩定性格傾向，
並有助於對各種情境採取目標策略。

Motives may "arouse or activate certain types of **goals**, but the goals then serve to guide and direct **behavior and achievement**" **(pintrich & Maehr, 2001, p.16).**
「動機」可以喚起或啟動某種類型的「目標」，但目標之後的作用是引導「行為」和「成就」。



Review / Motivational Goal Orientation

In the most recent research on **goal orientation**, two types of goals, **mastery** and **performance**, have been identified and each has both **approach** and **avoidance** valences **(Barnon, Finney, Davis & Owens, 2003; Elliot & McCregor, 2001; Pintrich & Maehr, 2001; Zuscho & Pintrich, 2000)**.
目標導向分為「精熟導向」和「表現導向」，且兩種類型都有「趨向導向」和「逃避導向」。

目標導向類型	趨向導向 approach	逃避導向 avoidance
精熟目標導向 mastery goal	趨向精熟目標導向 專注學習任務、自我成長	逃避精熟目標導向 避免誤解與犯錯
表現目標導向 performance goal	趨向表現目標導向 聚焦於優越感、好成績	逃避表現目標導向 避免看起來很笨、爛成績

資料來源：參考自《Goal Orientation Theories》(王淑玲，2024)

Review / Motivational Goal Orientation

Mastery approach goals describe students' desires to learn and understand, whereas the **avoidance element** portrays their hope to evade misunderstanding or completing a task incorrectly.
趨向精熟目標導向描述了學生學習和理解的渴望，而逃避精熟目標導向描述了他們希望避免誤解或錯誤完成任務。

Performance approach goals describe the desire to outdo others, whereas the **avoidance valence** focuses on evading the appearance of being less able than peers.
It has been shown **(Barron et al., 2003; Meece, Blumenfeld, & Hoyle, 1988; Pintrich, 2000)** that **more than one goal may be pursued simultaneously.**
趨向表現目標導向描述了學生想要超越他人的願望，而逃避表現目標導向則著重於迴避能力不如同儕的表象。而且研究顯示這類型的目標導向可以同時追求一個以上的目標。

Review / Motivational Goal Orientation

Pintrich and Maehr (2001) reported two studies:
Both studies used **self-report** measures to assess **motivational goal orientation**, **cognitive** and **metacognitive strategies**, and **related variables**.
這兩項研究都使用自我報告測量來評估動機目標導向、認知和後設認知策略，以及相關的變數。

(1) Performance approach goals had few negative and many positive outcomes,
whereas **performance avoidance goals** were related to poor consequences in both studies.
在這兩項研究中，趨向表現目標只有很少的負面結果和很多的正面結果，而逃避表現目標則會導致不良後果。

(2) Interest was positively related to **mastery approach goals** in both studies and negatively related to **mastery avoidance goals**;
(3) performance approach goals had a positive impact on **achievement** whereas that of **avoidance goals** was negative.
在兩項研究中，「興趣」與趨向精熟目標呈正相關，而與逃避精熟目標呈負相關；
趨向表現目標對「成就」有正面的影響，而逃避表現目標則是負面的。

Review / Motivational Goal Orientation

Pintrich and Maehr (2001) reported two studies:

(4) Both types of **avoidance goals** reported use of **surface strategies**.
兩種類型的「逃避導向」的學生都使用了「淺層的」學習策略。

(5) Students with **high approach tendencies (mastery and performance) used deeper elaboration and organizational strategies** than those high on avoidance.
有高度「趨向導向」的學生比有高度「逃避導向」的學生使用更深層的精緻化方法和組織型學習策略。

(6) Mastery approach goals were **positively related** to self-reported **metacognition**; similarly, and surprisingly, so were **all avoidance goals**.
趨向精熟目標導向與後設認知有正向關係;所有逃避目標導向也是如此。

(7) Both studies found that **avoidance goals** led to more **cognitive engagement** and greater use of **cognitive and metacognitive strategies** than did performance goals.
逃避目標導向比表現目標導向有更多的認知投入和後設認知策略的使用。

Review / Motivational Goal Orientation

Mastery goals have been associated with self-reported **intrinsic behavior (Elliot & Harackiewicz, 1996)** and greater use of both **cognitive** and **metacognitive strategies (Archer, 1994; Ford, Smith, Weissbein, Gully & Salas, 1998; Pintrich, 2000; Pintrich & Maehr, 2001 ; Schraw, Potenza & Nebelsick-Gullet, 1993)**.
精熟目標導向與內在動機行為和認知與後設認知策略的使用有關。

Performance goals typically led to **negative attributions**, use of **superficial strategies**, and **failure avoidance behavior**. Other researchers **(Ablard & Lipschultz, 1998; Arnes, 1992; Somuncuoglu & YiEdirim, 1999)** had similar findings.
表現目標導向通常會導致負面歸因、使用淺層的學習策略、做避免失敗的行為。
其他學者們也都有類似的發現。

Review / Motivational Goal Orientation

Mastery oriented students, compared to those with performance orientations, typically also use **more cognitive strategies**, have **greater engagement in and enjoyment** of learning activities, use **deeper learning processes**, and **seek more challenging tasks** than those with performance orientations.
精熟導向的學生通常會使用更多的認知策略，更投入和享受學習活動，更深層學習，並尋求更具挑戰性的任務。

Performance approach goals have a positive impact on learning.
Performance avoidance goals lead to evading activities that could make students appear to be incapable, to seeking easy problems, and to using surface rather than deep processing strategies.
趨向表現目標導向對學習有正面的影響。但逃避表現目標導向會導致學生迴避可能讓他們顯得能力不足的活動、尋求容易的問題，以及使用表面而非深層的處理策略。

Review / Motivation - Summary

WBL may be differentially effective for students with mastery orientations, who tend to work on content until it is comprehended, and use "deeper" cognitive, metacognitive and organizational strategies.
網路學習可能對精熟導向的學生有不同程度的效果，他們傾向於研究學習內容直到理解為止，並使用「更深層的」認知、後設認知和組織策略。

Performance avoidance oriented students tend to use shallower strategies, they compare themselves to their peers, and **tend to learn more in classroom contexts where exceeding peers is more salient.**
逃避表現導向的學生傾向於使用較淺的策略，他們會將自己與同儕比較，並傾向於在超越同儕更為突出的「教室」情境中學習。

03

Metacognition

導讀者：孫劍莉

Interest

- 興趣的影響**Interest's Impact**

Interest, or individual preferences for activities, influences learning, retention, and cognitive processing, with studies **(Elliott & Church, 1997; Elliot & McGregor, 2001, etc.)** showing a link between interest and mastery orientations.

興趣或個人對活動的偏好，會影響**學習、記憶與認知處理**，根據研究顯示，興趣與掌握取向之間存在關聯。

Interest

- 興趣的作用 Role of Interest

Interest enhances learning and recall, promoting deeper processing (Boekaerts & Boscolo, 2002); thus, adapting instruction to student interest can improve learning outcomes.

興趣有助於學習與回憶，促進更深層的處理**(Boekaerts & Boscolo, 2002)**；因此，**根據學生的興趣調整教學**可以**改善學習成效**。

興趣的類型　Types of Interest

興趣分為兩類：情境興趣（情境新奇或強烈引發）和主題興趣（對主題、任務或情境的持續偏好）

- Interest is classified into situational interest (elicited by novel or intense situations) and topic interest (enduring preference for topics, tasks, or contexts) (Renninger, Hidi, & Kxapp, 1992).

興趣與前知識的關係　Interest & Prior Knowledge

約**20%**的興趣差異與先前知識相關，顯示研究興趣時應考量前知識的影響。

- Around 20% of variance in interest is due to prior knowledge **(Tobias, 1994a),** indicating the importance of considering prior knowledge in interest research.

興趣與後設認知的研究　Research on Interest & Metacognition

興趣與後設認知的關聯研究有限。研究發現提高興趣的策略也提升了後設認知準確性。

- Limited research links interest to metacognition. Studies found that interest-enhancing strategies also improved metacognitive accuracy (Tobias, 1994b, 1995; McWhaw & Abrami, 2001).

後設認知的定義 Definition of Metacognition

後設認知是指監控、評估與規劃自己學習的能力

- Metacognition is defined as the ability to monitor, evaluate, and plan one's learning (Brown, 1980; Flavell, 1979).

後設認知的組成 Components of Metacognition

三個部分：對後設認知的知識、學習過程的監控與自我調節。

- Metacognition includes three components: knowledge about metacognition, monitoring of learning processes, and self-regulation **(Pintrich et al., 2000).**

層級後設認知過程 Hierarchical Metacognitive Process

後設認知可能是一個層級過程，先前學習的監控是更複雜控制過程的前提

- Metacognition may be a hierarchical process where monitoring prior learning is a prerequisite for more complex control processes **(Tobias & Everson, 2000).**

後設認知在自學環境中的重要性 Importance in Self-Directed Learning

在自學中，評估進度、選擇學習策略與計劃尤為重要，這些過程對於能區分已知與未知的學生更有優勢。

- In self-directed learning, progress evaluation, strategy selection, and planning are vital, with an advantage for students who can distinguish between the known and unknown.

後設認知的評估方法 Assessment Methods

後設認知通常通過觀察、訪談或自我報告量表來評估，但各方法有不同的挑戰

Metacognition is assessed through observation, interviews, or self-report scales, each presenting unique challenges **(Schraw & Impara, 2000).**

自我報告量表的挑戰 Challenges of Self-Report Scales

自我報告量表容易執行且評分簡單，但在評估動機或後設認知方面存在疑慮。例如，學生是否能真實描述學習中的動機與後設認知過程？

- Self-report scales are easily administered and scored but raise concerns when assessing motivation or metacognition. For instance, can students accurately describe and report their motivational and metacognitive processes during learning?

學業成績的優先性 Priority of Academic Achievement

研究發現，對於學生而言，獲得好成績是主要目標，而非對知識的掌握。學生在問卷中可能會選擇掌握目標，但實際上學業成績是更重要的動機。

- Research shows that students prioritize achieving good grades over mastering material **(Pressley et al., 1998).** They may select mastery goals on questionnaires, yet academic achievement remains the primary motivation.

後設認知與GPA的關聯 Metacognition and GPA Correlation

研究顯示，自我報告的動機策略學習問卷（MSLQ）及學習與學習技巧清單（LASSI）與GPA無顯著相關；而後設認知知識監控評估（KMA）與GPA有顯著關聯。

- Studies show no significant correlation between GPA and the Motivated Strategies for Learning Questionnaire (MSLQ) or Learning and Study Skills Inventory (LASSI); however, the Knowledge Monitoring Assessment (KMA) correlates significantly with GPA **(Tobias & Everson, 1998).**

自我報告與外部標準的比較 Self-Reports vs. External Standards

KMA與SAT成績的關聯比自我報告量表更高，顯示兩者測量的特徵不同

- The KMA showed a higher correlation with SAT scores than self-report scales, indicating that these assessments measure different **characteristics (Tobias et al., 1999; Pintrich et al., 2000).**

謹慎看待自我報告結果 Caution with Self-Report Results

這些結果顯示，自我報告在動機目標導向或後設認知上的結果應謹慎解讀。這些問題促使了更準確評估程序的發展。

- These findings suggest that self-report results on motivational goal orientation or metacognition should be interpreted cautiously, leading to the development of more reliable assessment methods.

後設認知知識監控評估（**KMA**） Knowledge Monitoring Assessment (KMA)

KMA可用於群體、電腦或網絡上，並具有客觀評分。學生先評估自己對知識的掌握程度，然後測試相同內容以比較準確性。

- The KMA can be administered to groups, via computers, or online, and is objectively scored. Students estimate their knowledge and are then tested on the same content to assess accuracy.

KMA與學業成績的關係 KMA and Academic Performance

研究顯示，知識估計越準確，後設認知越佳。**KMA**與小學生的閱讀與數學學習高度相關，與高職和大學生的成績呈中等相關

- Studies indicate that accurate knowledge estimates correlate with better metacognition. The KMA is strongly linked to reading and math in elementary students and moderately with grades in vocational and college students **(Tobias & Everson, 2000b, 2002)**

不同學生群體的KMA表現 KMA Performance in Different Groups

學習障礙和注意力缺陷的學生在知識監控準確度上低於普通學生，輟學生也低於在學學生。興趣提高準確度，而考試焦慮降低準確度。

- Students with learning disabilities or attention deficits show lower accuracy in knowledge monitoring than regular students, and dropouts perform lower than enrolled students. Interest improves accuracy, while test anxiety reduces it（**Gerrity & Tobias, 1996**）.

KMA在各學科中的應用 KMA Applications Across Subjects

KMA在各類學科（詞彙、數學、科學等）中皆顯示良好的結果，並與閱讀理解、數學成就、學習與GPA等有顯著的正相關。

- The KMA has shown consistent results across subjects (vocabulary, math, science, etc.) and positively correlates with reading comprehension, math achievement, learning, and GPA

KMA的優勢 Advantages of KMA

KMA不需要自我報告認知過程，減少了其他後設認知評估的限制，使其更為準確

- English: The KMA doesn't rely on self-reports of cognitive processes, reducing biases and making it more accurate compared to other metacognitive assessments **(Pintrich et al., 2000).**

04

Review / Help Seeking

導讀者：陳冠霖

背景

"Those who **sought help** were denigrated as **dependent**, and the act of relying on others was considered the antithesis of what it means to strive for excellence ....
That **view has changed** dramatically, however, with the recognition of **help seeking's** strategic value **in the learning process**" (Karabenick, 1998,p. 1).

尋求協助常被視為依賴他人、不獨立的象徵

近年來逐漸轉變,「尋求協助被視為能**幫助學習的行為**」

背景

Nelson Le Gall's research (1981) found that seeking help was an important skill and differentiated between two types of assistance. **Instrumental help** is the assistance (e,g., prompts, hints) necessary to complete tasks independently and has long-term strategic value by increasing future competence and independence. In contrast, **executive help** assists students in solving immediate problems but does not decrease their future dependence.

Nelson Le Gall's 將幫助方式分為兩種：

工具性幫助、執行性的幫助

背景

Nelson Le Gall's research (1981) found that seeking help was an important skill and differentiated between two types of assistance. **Instrumental help** is the assistance (e,g., prompts, hints) necessary to complete tasks **independently** and has **long-term** strategic value by increasing future competence and independence. In contrast, executive help assists students in solving immediate problems but does not decrease their future dependence.

工具性幫助:

提供學生**間接**的協助(例如:提示、暗示)

- 學生能**獨立**完成任務
- 具有**長期**提升能力的價值

背景

Nelson Le Gall's research (1981) found that seeking help was an important skill and differentiated between two types of assistance. Instrumental help is the assistance (e,g., prompts, hints) necessary to complete tasks independently and has long-term strategic value by increasing future competence and independence. In contrast, **executive help** assists students in solving **immediate problems** but does not decrease their future **dependence**.

執行性幫助：

提供學生**直接**的協助（例如：答案）

- 學生較**沒有學習到獨立學習能力**

背景

Most help seeking research was conducted in classrooms although **Aleven. Stahl, Schworm, Fischer, and Wallace (2003)** reviewed a number of such studies in interactive learning environments, including WBL. Their review suggests that "an increasing number of studies provide evidence that learners often **do not use** help functions very effectively or even **ignore them** totally"(p. 278).

Aleven. Stahl, Schworm, Fischer, and Wallace (2003)的研究指出

- 學習者常常**不善於使用求助技能**，甚至完全不主動求助。

教室中的求助技能研究

In classroom, help seeking research has been conducted largely with **self-report scales.** Some study (e.g., Butler, 1998) used both **self-reports and observations**; others were based largely on **observation**—although these usually occurred in elementary schools (e.g., Nelson-Le Gall & Jones, 1990).

研究對象大多是國小階段:

- **自陳式量表(最多)**
- **自陳式量表與觀察(第二)**
- **觀察(最少)**

成就目標與求助技能

Results indicate (Karabenick, 1998) that **mastery students** seek more needed assistance whereas students with **performance avoidance** orientations often do not seek help fearing that it makes them seem incompetent.

趨向精熟的學生:**願意**尋求協助

逃避表現的學生:**不願意**尋求協助

成就目標與求助技能

Results indicate (Arbreton, 1998) that **mastery oriented students** seek more help, especially **instrumental assistance**, than their **performance oriented peers** who, when they do seek it, are more likely to look for **executive help**.

趨向精熟的學生：工具性幫助

逃避表現的學生：執行性幫助

求助技能的兩個主要階段及其研究限制

Seeking assistance is composed of two hierarchical components. First, students must recognize making a **mistake**, having a **misconception**, or hitting a **learning impasse**. Second, they must try to **overcome the problem with help**.

Virtually all help seeking research has investigated only the motivational component; **awareness of error** has received little explicit study.

第一步：**識別錯誤、迷思** 或**學習困難**。

第二步：**尋求幫助**解決問題。

＃研究著重於動機，較少**覺察錯誤（認知層面）**的相關研究。

KMA理論與求助技能的關聯性研究

Romero and Tobias (1996) found that **accurate monitors** sought more needed help, that is, **on words estimated to be unknown. Less accurate students** asked for more help when it was not needed, that is, **on items estimated as known**.

知識監控較準確（後設認知能力高）的學生:**未知詞彙**（必要幫助）

知識監控較不準確（後設認知能力低）的學生:**已知詞彙**（不必要的幫助）

KMA理論與求助技能的關聯性研究

In a succeeding study using math problems (Tobias & Everson, 1998) **accurate knowledge monitors** sought more help than others on two types of items where it seemed to be needed; those originally **estimated as soluble** and **failed on the test**; here, help probably assisted students to find the reasons for wrong estimates; and those **estimated to be insoluble and failed on the test.**

知識監控較準確（後設認知能力高）的學生：

會寫但答錯的題目、不會寫且答錯的題目

結論建議

Most of the **KMA help seeking studies** ignored the **motivational component**, whereas **help seeking research in classrooms** has ignored the **metacognitive component**. Clearly both variables should be studied in order to have an accurate picture of students' help seeking.

KMA理論與求助技能的關聯性研究：容易**忽略動機層面**

教室中的求助技能研究：容易**忽略後設認知層面**

建議求助技能的研究要**同時考量**動機與後設認知層面

統整

	求助技能好	求助技能不好
目標取向	趨向精熟	逃避表現
動機	願意求助	不願意求助
後設認知能力	好	不好
問問題技巧	問關鍵的問題	問無關的問題

05

Suggestions For Further Research

導讀者:柯廷龍

Suggestions For Further Research

The findings document the importance of metacognitive and motivational variables for learning generally and for WBL specifically, although very few studies examined both variables simultaneously other than by self-report scales.

即使有對於後設認知和動機如何影響學習和網路學習相關的許多文獻，但除了以自我檢測量表作實驗的研究以外，上述**兩個變項通常都不會放在一起研究**。

Examination of the joint effects of both variables is especially important in WBL where knowing why individuals chose to receive instruction over the Internet, and understanding the variables leading to success are equally important.

了解學生選擇網路學習的**原因**和導致學習**成功的變項**都一樣重要。

Suggestions For Further Research

Investigating both motivational and metacognitive variables in studies of seeking help is especially important because the metacognitive recognition of a learning impasse, not generally investigated, must precede the desire to overcome the problem.

在尋求協助的研究中探討動機和後設認知變數很重要，因為會**先藉由後設認知界定學習問題後才會有渴望去克服它。**

Suggestions For Further Research

- 網路學習研究的缺乏

There are few studies of metacognition or motivation in WBL, and they often use questionnaires developed for that investigation that have unknown construct validity, and often unknown reliability as well.

在網路學習中後設認知和動機的學習研究都很少，這些研究都使用問卷調查，而大部分這些研究的問件調查的**信度和效度都讓人質疑**。

Suggestions For Further Research

Motivational research in WBL would profit from studies employing the motivational goal orientation perspective, especially because the scales to assess it are widely used and their construct validity and reliability are well known (Barron et al., 2003;
Pastor, Barron, Miller, & Davis, 2004).

WBL 可以從使用動機導向的研究獲益，特別是因為用來**評估的量表被廣泛使用**，並且它們的**結構效度和信度是眾所周知的**。(Barron et al., 2003; Pastor, Barron, Miller, & Davis, 2004).

Suggestions For Further Research

- 動機與網路學習研究

Motivational orientation seems especially important in students‘ learning from the Internet because they often work in isolation without the social support of instructors or fellow students. In WBL, students may feel more anonymity than they do in classes, hence performance oriented students should feel that seeking assistance poses little threat to their self-esteem,
a finding reported by Schofield (1995) in computer-based learning environments.

動機對於網路學習很重要，因為他們通常都必須在一個**獨立、沒有從指導員來的社會支持或沒有同學的情境**下學習。Schofield(1995) 指出因為網路學習在較為匿名的環境下，表現導向的學生應該會覺得尋求幫助對他們的**自尊構成些微的威脅**。

Suggestions For Further Research

- 後設認知在網路學習研究的缺乏

Metacognition has been investigated even less frequently in WBL than motivation. That scarcity is also surprising because students‘ ability to organize and evaluate what they have learned, and their selection of effective strategies are likely to be more important in WBL than during face-to-face instruction where students can turn to both instructors and to peers for help.

後設認知在網路學習的研究比動機更少。這點令人震驚，因為學生在網路學習情境中**組織和評估所學，以及選擇有效策略的技能可能比面對面學習更重要**。在面對面學習的情境中，學生可以向老師和同學請教問題。

Suggestions For Further Research

- 先備知識與網路學習研究

Understanding learning, especially in WBL, probably also requires consideration of the effects of students' prior knowledge(Shapiro, 2004), in addition to motivation and metacognition. Furthermore, clarification of students' help seeking during instruction, and the types of help sought are probably also affected by their prior knowledge.

了解學習，尤其是網路學習，除了動機和後設認知以外，可能也要**考慮學生的先備知識**。學生在學習時尋求協助的行為和協助的種類有可能也會被他們的先備知識影響。

Suggestions For Further Research

- 先備知識與網路學習研究

Aleven, et al. (2003) concluded that students "with low prior knowledge—those who need help the most—are least likely touse help appropriately when help is under student control" (p. 298).

Aleven, et al. (2003) 認為先備知識較少的學生，也就是那些**最需要幫助的學生，比較不可能適當使用**在他們手上的學習輔助資源。(p. 298).

Suggestions For Further Research

- 先備知識與網路學習研究

It has been suggested (Tobias, 1989a) that such students need instructional support most in order to succeed, but Aleven, et al's. review suggests that they are least likely to seek it voluntarily.

有學者建議，缺乏先備知識的學生較需要教學支持才能成功，但Aleven, etal's. 指出這些學生比較**不會自願尋求這些支持**。

Suggestions For Further Research

- 先備知識與網路學習研究

In a computer assisted instructional context, it was found (Tobias, 1989b) that students with comprehension problems and limited prior knowledge performed optimally when they were required to review their reading. Perhaps studies varying optional and mandated help for students with different levels of prior domain knowledge may also be helpful in WBL.

在一個電腦輔助學習的情境，研究發現(Tobias, 1989b)有理解問題和先備知識匱乏的學生在**被要求要複習的時候表現最好**。或許研究將**選擇性和強迫性學習輔助**為變數用於具有不同先備知識等級的學生在網路學習情境會有幫助。

Suggestions For Further Research

Fletcher and Tobias (in press) also document the importance of prior knowledge in multimedia learning environments

Fletcher and Tobias (尚未出版) 也記載了先備知識在多媒體學習情境的重要性

Suggestions For Further Research

- 先備知識與動機和學生尋求幫助的關係

Findings that students with limited prior knowledge seek help less often than others seem similar to the results relating goal orientation to help seeking. Because mastery oriented students typically learn more than their performance oriented peers, they are more likely to have high prior knowledge and seek more help.

尋求幫助的研究在目標導向的研究也有類似學生先備知識尋求幫助的研究相似的結果。因為**精熟導向**的學生通常會學得比表現導向的同學更多，所以他們也比較可能會有較多的先備知識且尋求更多的幫助。

Suggestions For Further Research

- 興趣與學生尋求幫助的關係

An intriguing finding reported by Dochy, Segers, and Buehl (1999) is that high interest can compensate for low prior knowledge, perhaps because highly interested students may work harder than those with lower interest, a finding that should also be investigated in WBL contexts.

Dochy, Segers, and Buehl (1999) 發現**高度興趣可以彌補先備知識的不足**，有可能是因為有興趣的學生會比那些較不感興趣更努力學習，這個發現也應該在網路學習的情境去探討。

Suggestions For Further Research

- 結論

The results point to the importance of examining interest, metacognition, goal orientation, and prior knowledge simultaneously in studies dealing with student characteristics and WBL.

上述總結就是在研究學生特質和網路學習情境中同時探討興趣、後設認知、目標導向和先備知識這些變數很重要。

Suggestions For Further Research

- 結論

Such research may clarify both the variables accounting for success in WBL and make it possible to identify the types of students who may profit maximally from Internet-based instruction. Furthermore, research combining these variables is equally scarce in studies of instruction generally and, therefore, Will also clarify the effects of the interaction of these variables on the outcomes of learning in varying contexts.

上述提到的研究可以釐清這些變項對於網路學習的成效，可能可以找出哪些類別的學生可以從網路學習情境獲益最多。另外，在包含所有學習情境中有結合這些變項的研究本來就很少，所以上述提到的研究也可以釐清這些變相互動在不同情境中的結果。

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Tobias, S. (2013). The importance of motivation,metacognition, and help seeking in web-based learning. In *Web-Based Learning* (pp. 203-220). Routledge.